STUDY OVERVIEW

This study quantifies how stress builds and propagates across AAPL, MSFT, GOOGL, AMZN, META, and NVDA over five years of daily data (April 2021–April 2026, 1,305 trading days). K-Means clustering [MacQueen 1967] on six rolling state features segments history into three distinct market regimes. A custom Stress Propagation Score tracks contagion in real time; ARIMA(2,0,1) [Box & Jenkins 1970] forecasts realized volatility on an expanding walk-forward; and GBM Monte Carlo [Samuelson 1965; Rockafellar & Uryasev 2000] estimates VaR and CVaR. Key result: Stress Contagion averaged 45.9% annualized vol versus 7.7% in Calm Expansion — a six-fold difference — with correlation rising from 0.14 to 0.82.

DATASET

Universe:	AAPL, MSFT, GOOGL, AMZN, META, NVDA
Period:	2021-04-18 – 2026-04-18
Frequency:	Daily adjusted-close OHLCV
Portfolio:	Equal-weight synthetic portfolio
Source:	Yahoo Finance via yfinance (pre-downloaded)
Trading days:	1,305 after alignment & forward-fill
Preprocessing:	Forward-fill ≤2 gaps; drop remaining NaN

ANALYTICAL METHODS

Rolling features:	Realized vol, avg pairwise corr [Engle 1982], breadth above 50-DMA, dispersion, lead-lag concentration
Stress Score:	Equal-weight z-score blend of the 5 metrics; also the 6th K-Means clustering feature
Regime detection:	K-Means k=3, n_init=25 [MacQueen 1967] on the 6-feature rolling state matrix
Stationarity:	Augmented Dickey-Fuller [Dickey & Fuller 1979] on prices, log-returns, vol, and stress score
Autocorrelation:	Ljung-Box Q-test [Ljung & Box 1978] on log-returns (efficiency) and ARIMA residuals
ARIMA(2,0,1):	20-day realized vol forecast; AIC/BIC grid search; expanding walk-forward [Box & Jenkins 1970]
GBM Monte Carlo:	500 paths, 252-day horizon; VaR 95/99% and CVaR 95/99% [Rockafellar & Uryasev 2000]

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Dickey & Fuller (1979). JASA, 74(366), 427–431.

Engle, R.F. (1982). Econometrica, 50(4), 987–1007.

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MARKET REGIME SUMMARY

Regime	Days	% of Sample	Avg Vol (20d ann.)	Avg Corr (20d)	Avg Breadth	Avg Stress Score
Calm Expansion	514	40.0%	7.6%	0.14	0.63	-0.93
Rotation / Transition	501	39.0%	17.9%	0.37	0.54	+0.18
Stress Contagion	270	21.0%	45.9%	0.82	0.14	+1.49

REGIME INTERPRETATION

Calm Expansion

Vol 7.6% Corr 0.14 Breadth 0.63 N=514 (40%)

Corr 0.14: stocks move on fundamentals [Engle 1982]. Self-sustaining at 99.8% daily persistence [MacQueen 1967].

Rotation / Transition

Vol 17.9% Corr 0.37 Breadth 0.54 N=501 (39%)

Corr rises (0.37), breadth falls (0.54). Gating regime to Stress: 2.8% daily escalation probability.

Stress Contagion

Vol 45.9% Corr 0.82 Breadth 0.14 N=270 (21%)

Lock-step moves (corr 0.82) eliminate diversification. Breadth 0.14. Self-sustaining at 95.2% daily [Rockafellar & Uryasev 2000].

Fig. 1 — Stress Propagation Score (green = Calm, amber = Rotation, red = Stress)

ARIMA VOLATILITY FORECASTING

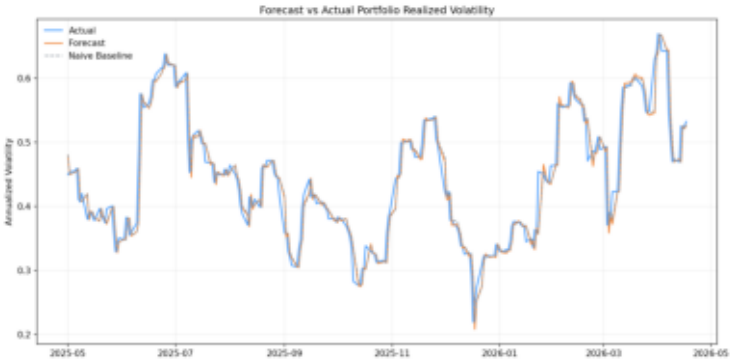
Realized vol is stationary (ADF $p < 0.001$) [Dickey & Fuller 1979]. AIC/BIC grid search selected ARIMA(2,0,1) [Box & Jenkins 1970]. Walk-forward RMSE 0.0269 vs naive 0.0271 — near-parity reflects ARCH-like vol persistence [Engle 1982]. Model value: structural validation via AIC -6592.9, BIC -6568.2, and white-noise residuals [Ljung & Box 1978] ($p > 0.05$).

KEY FINDINGS

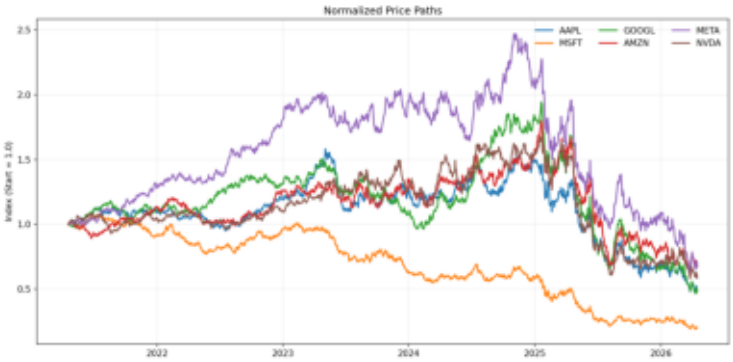
- Regime separation: Stress Contagion averaged 45.9% realized vol versus 7.6% Calm — a 6.0× difference. Correlation rose 0.14→0.82.
- Stress Propagation Score: +1.49 in Stress, -0.93 in Calm — a 2.4σ separation confirming strong real-time regime classification power.
- Score rose before only 35.7% of stress entries: its value is real-time state classification, not advance prediction — consistent with abrupt regime shifts that preclude early warning.
- ARIMA(2,0,1) RMSE 0.0269 ≈ naive baseline 0.0271. Near-parity is expected: realized vol is near-random-walk. Structural validation (AIC/BIC, white-noise residuals) is the meaningful result.

Stress Propagation Score & Regime Shading

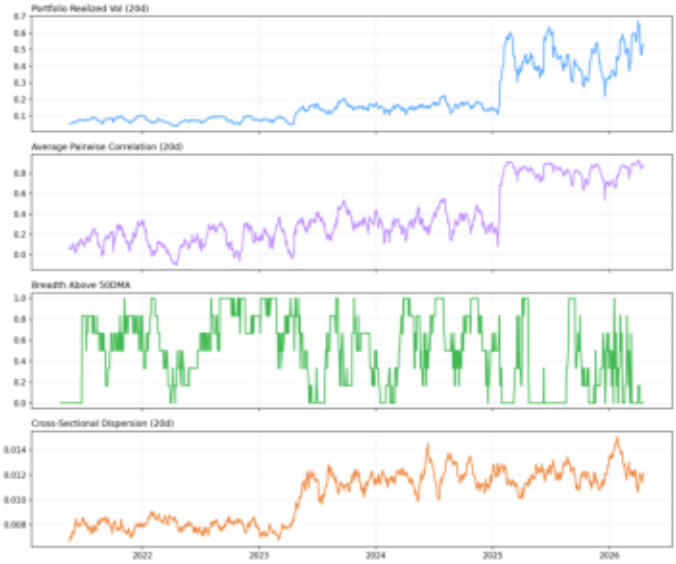
ARIMA(2,0,1): Walk-Forward Forecast vs Actual



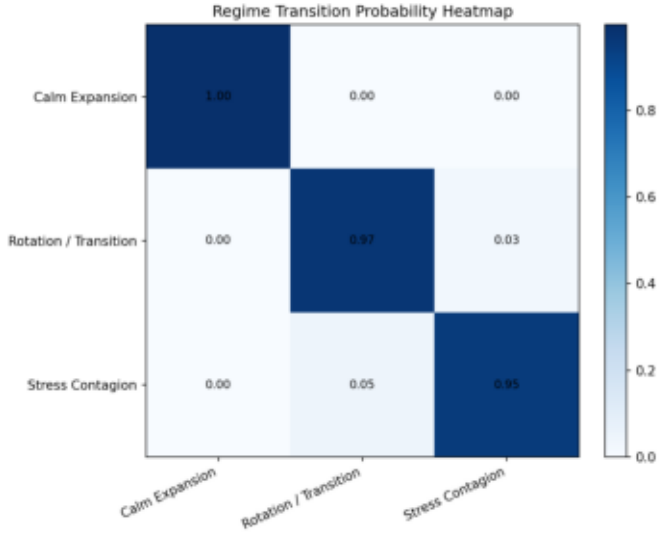
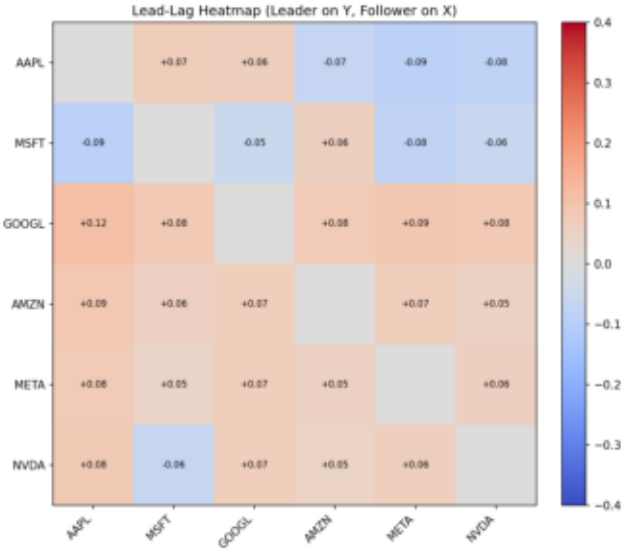
Normalized Price Paths (Base = 1.0)



Rolling Portfolio State Metrics (20d window)



Regime Transition Probability Heatmap



Lead-Lag Heatmap (Leader on Y-axis)